An Overview on Reed-Muller, Polar, and Automorphism Ensemble Decoding

Pin-Jing, Li

ouo.ee11@nycu.edu.tw
National Yang Ming Chiao Tung University

August 18, 2025

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Recursive Projection-Aggregation (RPA) Decoding of RM code

- **Input:** Received word $y \in \mathbb{R}^{2^m}$ from a noisy channel.
- Choose k nonzero directions $a_1, \ldots, a_k \in \mathbb{F}_2^m$.
- For each direction a_i:
  - Define subspace $\mathbb{B}_i = \{x \in \mathbb{F}_2^m : \langle a_i, x \rangle = 0\}$.
  - Form cosets $\mathbb{B}_i^{(0)} = \mathbb{B}_i$, $\mathbb{B}_i^{(1)} = \mathbb{B}_i + a_i$
  - Restrict y to each coset: $y^{(0)} = y|_{\mathbb{B}_i^{(0)}}, \quad y^{(1)} = y|_{\mathbb{B}_i^{(1)}} \in \mathbb{R}^{2^{m-1}}$
  - Recursively decode as $\mathrm{RM}(m-1, r)$, get $\hat{c}^{(i)} \in \{0,1\}^{2^m}$.
- Aggregate results over all k directions $\hat{c}_j = \text{majority} \left\{ \hat{c}_j^{(i)} \right\}_{i=1}^{k}$
- **Output:** Final decoded codeword $\hat{c} \in \{0,1\}^{2^m}$

Automorphism Ensemble Decoding (AED) of RM code

- **Input:** Received soft vector $y \in \mathbb{R}^N$ and code $\mathcal{C} \subseteq \{0,1\}^N$ with automorphism group $\mathrm{Aut}(\mathcal{C})$
- **1. Select automorphisms:** Permutation $\mathbf{x} = \mathbf{Az} + \mathbf{b}$ is an automorphic actions on $\mathcal{C}$

$$\Pi = \{\pi_1, \ldots, \pi_k\} \subseteq \mathrm{Aut}(\mathcal{C}), \quad \pi_i : [N] \to [N]$$

- **2. Permute received vector:** For each i, let $y^{(i)} = \pi_i(y)$
- **3. Decode:** Apply a base decoder D to obtain $\hat{c}^{(i)} = D(y^{(i)})$
- **4. Invert permutation:** $\tilde{c}^{(i)} = \pi_i^{-1}(\hat{c}^{(i)})$
- **5. Aggregate:** Final decoded codeword by majority vote:

$$\hat{c}_j = \mathrm{majority}(\tilde{c}_j^{(1)}, \ldots, \tilde{c}_j^{(k)}), \quad \forall j \in [N]$$

UTL Automorphisms (Phillet) (1)

Goal: Given a UTL matrix $A \in \mathrm{UTL}_n$, construct a monomial set $\mathcal{F} \subseteq \mathcal{M}_n$ such that:

$$A(\mathcal{F}) = \mathcal{F}$$

that the code is invariant under the transformation $\mathbf{x}' = A\mathbf{x}$

Monomial Code

- Let $N = 2^n$, each index $i \in [N]$ maps to binary vector $\mathbf{i} \in \mathbb{F}_2^n$
- Associated monomial: $M_{\mathbf{i}}(x_1, \ldots, x_n) = \prod_{j=1}^{n} x_j^{i_j}$

UTL

- Let $A \in \mathrm{UTL}_n$ be upper-triangular with ones on the diagonal
- Define transformation $T : \mathbf{x} \mapsto A\mathbf{x}$
- Automorphism condition: $M \circ T \in \mathcal{F} \quad \forall M \in \mathcal{F}$

UTL Automorphisms (Phillet) (2)

Variable Substitution

- Each $A_{i,j} = 1$ implies substitution $x_i \mapsto x_i + x_j$
- Affects all monomials containing x_i
- Transformed monomial: $M'(x) = M(x_1, \ldots, x_i + x_j, \ldots, x_n)$

Feasibility Check

- Define $\mathcal{F}_{i,j} := \{M' \mid M \in \mathcal{F}, x_i \mapsto x_i + x_j\}$
- If $\mathcal{F}_{i,j} \subseteq \mathcal{F}$, then $A_{i,j}$ is admissible
- Otherwise, identify missing monomials $\mathcal{M}_{\text{add}} \subseteq \mathcal{F}_{i,j} \setminus \mathcal{F}$

Monomial Set Augmentation

- Update: $\mathcal{F} \leftarrow \mathcal{F} \cup \mathcal{M}_{\text{add}}$
- If dimension K must be fixed, remove least reliable monomials using reliability sequence $\mathcal{R}$

- best set of permutation
- bounds on AED performance relative to ML decoding
- Creating algebraic structure on probability-selected generating monomials